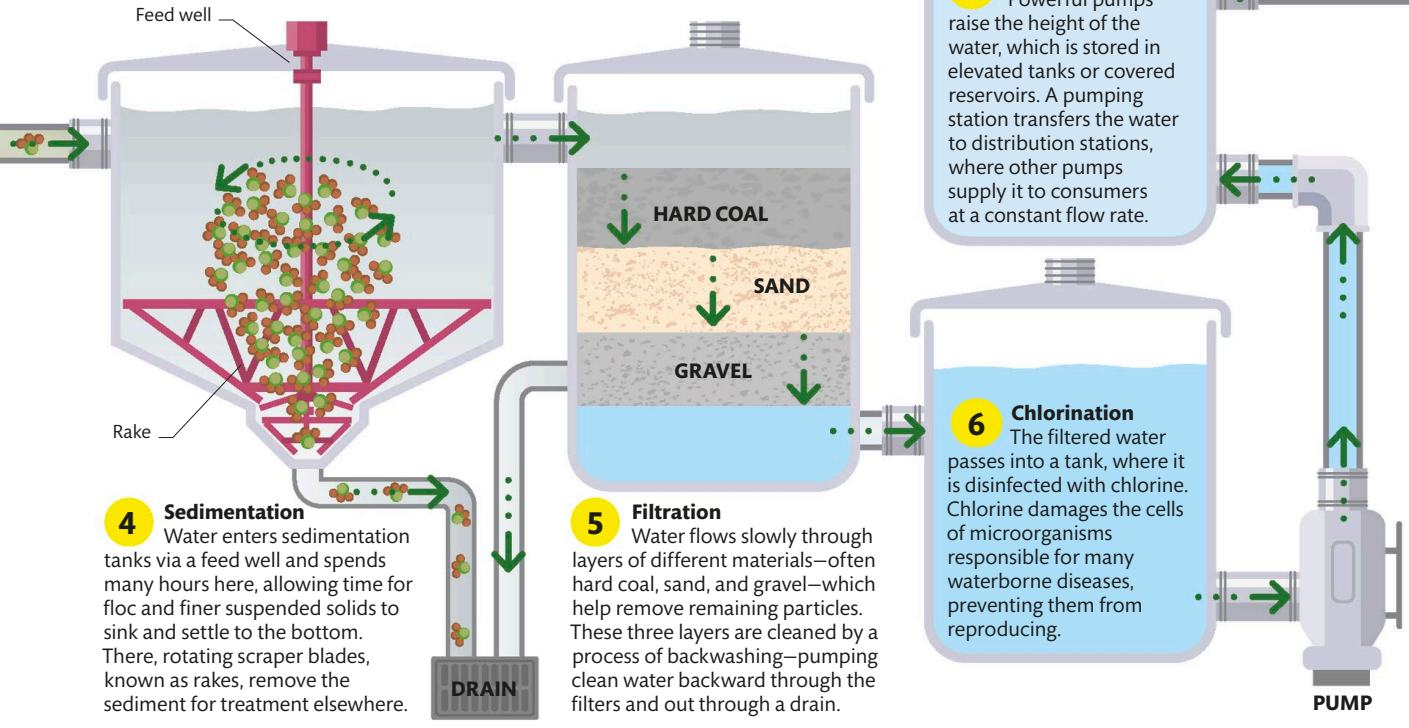
844 MILLION

THE NUMBER OF **PEOPLE** WHO LACK ACCESS TO CLEAN **DRINKING WATER**

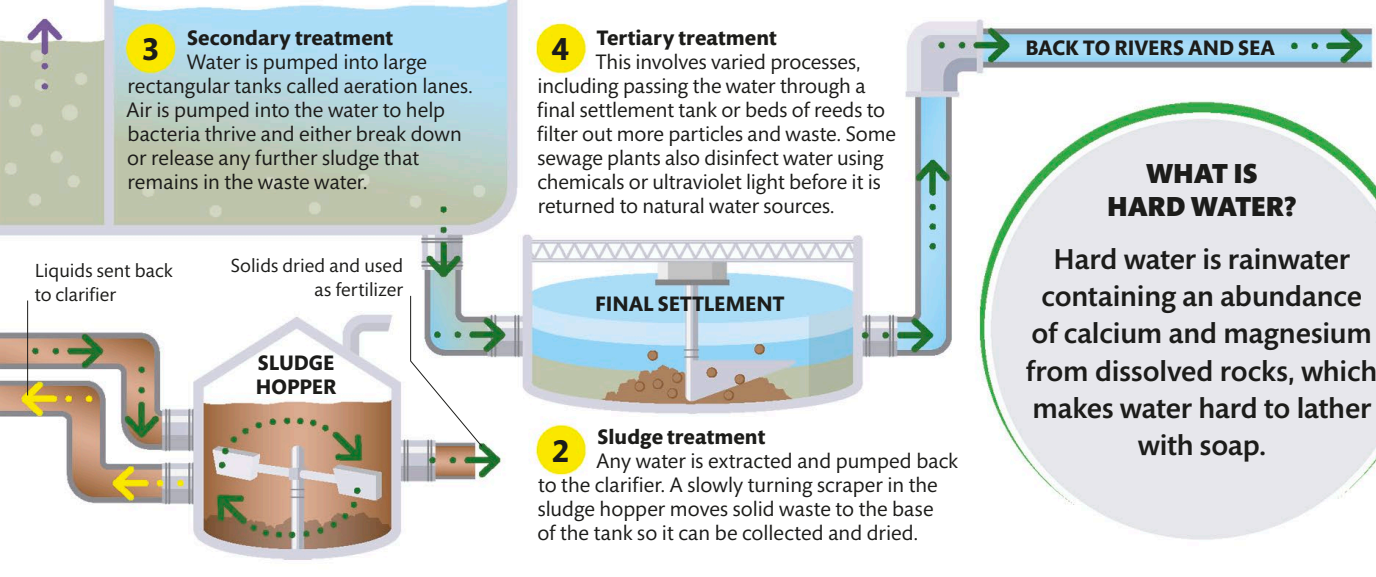


4 Sedimentation
Water enters sedimentation tanks via a feed well and spends many hours here, allowing time for floc and finer suspended solids to sink and settle to the bottom. There, rotating scraper blades, known as rakes, remove the sediment for treatment elsewhere.

5 Filtration
Water flows slowly through layers of different materials—often hard coal, sand, and gravel—which help remove remaining particles. These three layers are cleaned by a process of backwashing—pumping clean water backward through the filters and out through a drain.

6 Chlorination
The filtered water passes into a tank, where it is disinfected with chlorine. Chlorine damages the cells of microorganisms responsible for many waterborne diseases, preventing them from reproducing.

7 Storage
Powerful pumps raise the height of the water, which is stored in elevated tanks or covered reservoirs. A pumping station transfers the water to distribution stations, where other pumps supply it to consumers at a constant flow rate.



3 Secondary treatment
Water is pumped into large rectangular tanks called aeration lanes. Air is pumped into the water to help bacteria thrive and either break down or release any further sludge that remains in the waste water.

4 Tertiary treatment
This involves varied processes, including passing the water through a final settlement tank or beds of reeds to filter out more particles and waste. Some sewage plants also disinfect water using chemicals or ultraviolet light before it is returned to natural water sources.

2 Sludge treatment
Any water is extracted and pumped back to the clarifier. A slowly turning scraper in the sludge hopper moves solid waste to the base of the tank so it can be collected and dried.

WHAT IS HARD WATER?

Hard water is rainwater containing an abundance of calcium and magnesium from dissolved rocks, which makes water hard to lather with soap.